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Department of Mechatronics Engineering



Transducers and Sensors Lab (0908473)

Course Project: Solar Tracker Using Light Dependent Resistance (LDR)

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Objectives

This project aims to showcase our understanding of the principles, construction, and characteristics of light dependent resistance (LDR), as well as the transduction circuits used alongside it. We will also be utilizing it in a real-life application, specifically the sun tracking solar cell.

Introduction

Renewable energy has gained a huge amount of attention in recent years. Being a cleaner, environmentally friendly alternative to other means of energy production, solar energy has become one of the most used means of energy production. Nowadays, solar cells can be found on almost every roof. To increase their efficiency, a suggested method is mobilizing the solar cells in a way that maximizes the sunlight falling upon them, allowing them to collect as much energy as possible. This requires the usage of a sensor or transducer that can detect light efficiently, such as light dependent resistance (LDR).

Theory

A light dependent resistor, or LDR for short, is a nonlinear transducer that is highly sensitive to light. It is also referred to using the terms photoresistor, photoconductor, and CdS device as it is a resistance made up of a semiconducting material called cadmium sulfide. Unlike typical semiconductors, which use silicon doped with other elements to increase conductivity, CdS has high electrical resistance in the dark. When it is subjected to light, photons get absorbed by the material, providing its valence electrons with enough energy to move into the conduction band, creating electron-hole pairs with free electrons that increase conductivity, directly decreasing the CdS resistance.

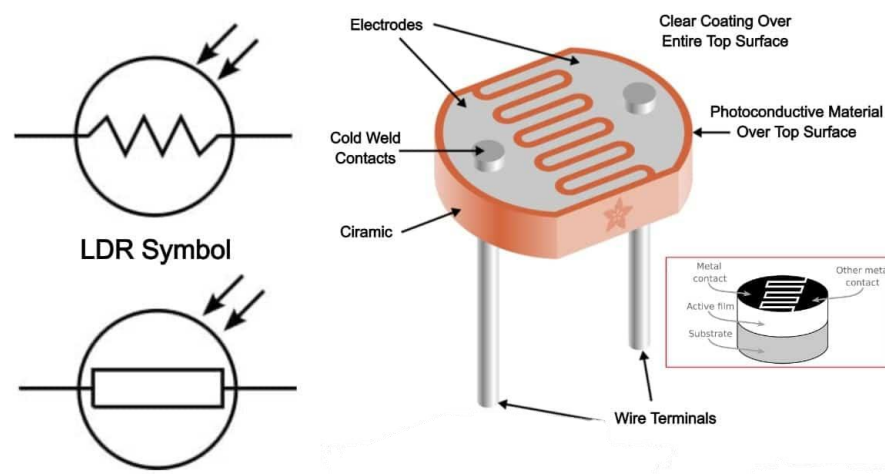


Figure 1: The structure of an LDR

Often used in automated applications, LDRs are found in most street lamps, solar garden lights, and even brightness sensors in electronic devices. In addition, many alarm systems utilize LDRs to detect changes in lighting.

LDR Working Principle

As mentioned before, an LDR is made up of the light sensitive semiconductor CdS, meaning it operates using the principle of photoconductivity. When light is shone on a photoresistor, its resistance decreases, allowing current to flow through. The amount the resistance drops is related to the light level, or illuminance. Illuminance is known as the sum of luminous flux on a surface per unit area. It can be measured in different units such as foot candles (lumen/ft²) and lux (lumen/m²). This means that an LDR's resistance will be at its highest in a completely dark environment.

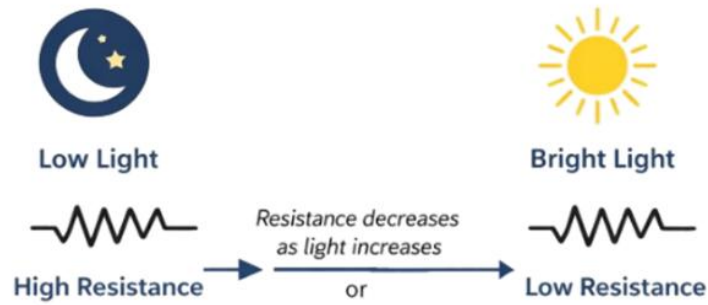


Figure 2: Working principle of LDR

LDR Signal Conditioning

To extract useable data from an LDR, we must convert its resistance to voltage or some other type of electrical signal that can be read by microcontrollers and ADCs. To do so, we use a signal conditioning circuit that consists of a DC voltage source, a fixed resistance, and the LDR in voltage divider configuration as shown in figure 3.

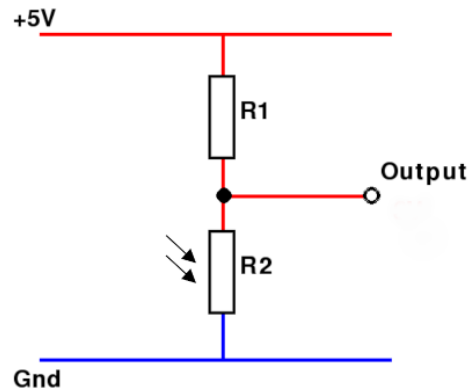


Figure 3: LDR conditioning circuit

With a simple circuit like this, filtering and amplification are usually not required. The general output formula is:

$$V_{out} = V_{cc} \times \frac{R_2}{R_2 + R_1}$$

Methodology

Phase One: System Design and Workflow

For simplicity, the system is designed as a single-axis tracker, requiring two data inputs from both sides of the axis, one microcontroller, and one actuator to move the solar cell accordingly. The LDRs are responsible for detecting the different amounts of light shining from different directions, and an output voltage from each side will be then fed into the microcontroller, which then compares their values and controls an actuator to move in the direction with the most light/highest voltage detected.

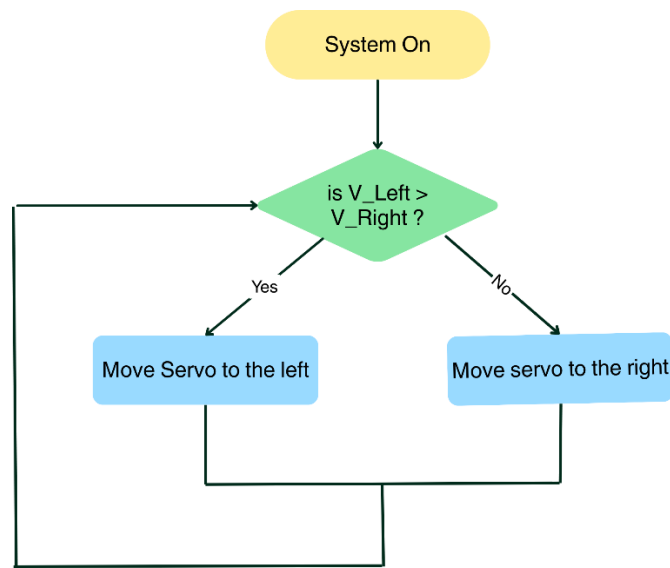


Figure 4: System flowchart.

Phase Two: System Hardware and Signal Conditioning

To provide output voltages usable by the microcontroller, the system requires a signal conditioning circuit alongside the LDRs. We used a voltage divider circuit that utilizes the microcontroller's 5V output and two 10kΩ resistors which we found to be the most suitable for the LDR's working conditions, providing high sensitivity to light. This results in the following output voltage formula:

$$V_{out} = 5V \times \frac{10k\Omega}{10k\Omega + LDR}$$

An Arduino Nano was used as a microcontroller for its compact size, and a servo motor for the actuator since it is easy to control. The solar cell frame (see figure 5) is tilted at a 27° angle (standard local solar panel angle) and includes a notch between the LDRs to create a shadow that is meant to enhance the difference between readings and allow for better accuracy and a stronger sense of direction.

Below is a list of all the equipment and devices used to build the system:

- 2 LDRs (5mm, 50.0-1.0M Ω). [1]
- Arduino Nano. [2]
- MG996R servo motor. [3]
- Resistors (10k Ω).
- Batteries (6V).
- Solar cell. [4]
- Solar cell frame.

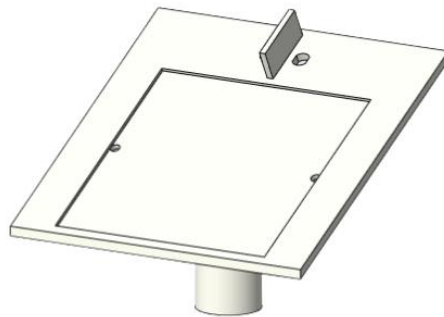


Figure 5: Solar cell frame designed for this project.

The signal conditioning circuit for the LDRs is connected as follows:

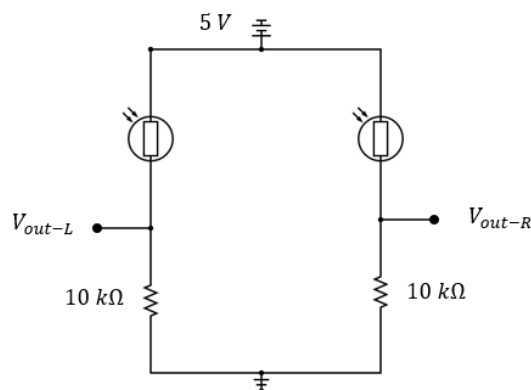


Figure 6: System signal conditioning circuit.

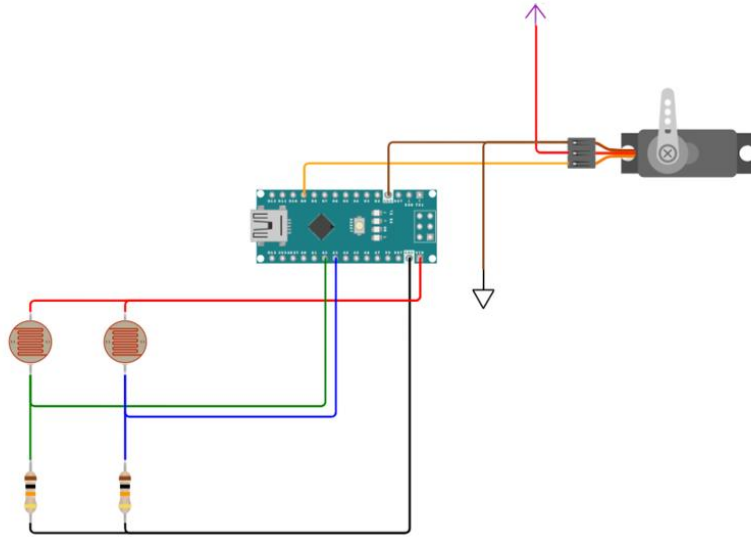


Figure 7: Complete system circuit diagram.

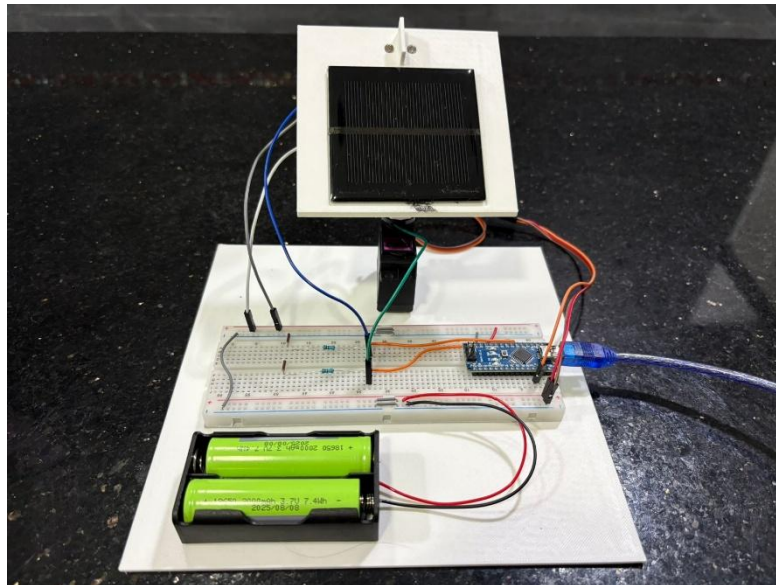


Figure 8: Real life system build.

Phase Three: System Software

To apply the system's logic, we supplied the Arduino Nano with the following code:

```
#include <Servo.h>

Servo myservo;
int pos = 90;
int tolerance = 10;

void setup() {
  Serial.begin(9600);
  myservo.attach(9, 500, 2400);
}

void loop() {
  int Vout_L = analogRead(A2);
  int Vout_R = analogRead(A3);

  // Convert to voltage
  float voltageL = (Vout_L * 5.0) / 1023.0;
  float voltageR = (Vout_R * 5.0) / 1023.0;

  Serial.print(voltageL);
  Serial.print(",");
  Serial.println(voltageR);

  // System Workflow
  if (Vout_R > (Vout_L + tolerance)) {
    if (pos < 175) pos += 2;
  }
  else if (Vout_L > (Vout_R + tolerance)) {
    if (pos > 5) pos -= 2;
  }

  myservo.write(pos);
  delay(50);
}
```

Data Monitoring

Using the Arduino Nano's collected data, we can observe the change in the output voltages and calculate the LDRs resistances using the following MATLAB code:

```
% 1. Setup Serial Port
arduinoPort = serialport("COM3", 9600);
configureTerminator(arduinoPort, "LF");
flush(arduinoPort);

% Constants for calculation
Vcc = 5;
R_fixed = 10000; % 10k Ohm resistor
maxPoints = 100; % Number of data points to show on screen at once

% 2. Initialize Tracking Counters
timeCounter = 0;

% 3. Setup Figure Window
figureHandle = figure('Name', 'LDR Real-Time Dual Analytics', 'NumberTitle', 'off');
clf;

% --- Voltage Graph ---
subplot(2, 1, 1); % 2 rows, 1 column, graph #1
lineVoltLeft = animatedline('Color', 'b', 'LineWidth', 2, 'DisplayName', 'Left V');
lineVoltRight = animatedline('Color', 'r', 'LineWidth', 2, 'DisplayName', 'Right V');
ylabel('Voltage (V)');
title('Real-Time Sensor Voltages');
ylim([0, 5.2]); % Pin the voltage axis between 0V and 5V
grid on;
legend('Location', 'northwest');

% --- Resistance Graph ---
subplot(2, 1, 2);
lineResLeft = animatedline('Color', 'b', 'LineWidth', 2, 'DisplayName', 'Left LDR');
lineResRight = animatedline('Color', 'r', 'LineWidth', 2, 'DisplayName', 'Right LDR');
xlabel('Time / Data Points');
ylabel('Resistance (Ohms)');
title('Real-Time LDR Resistances');
grid on;
legend('Location', 'northwest');

disp('Plotting live voltage and resistance charts... Close the window to stop.');
```

```
% 4. Main Processing Loop
while ishandle(figureHandle)
    try
        rawData = readline(arduinoPort);
        dataArray = str2double(split(rawData, ","));

        if numel(dataArray) == 2
            % Accept the voltages directly from Arduino
            Vout_L = dataArray(1);
            Vout_R = dataArray(2);
```

```

% Calculate Resistances (with dark-threshold safety checks)
if Vout_L > 0.05
    R_LDR_Left = ((R_fixed * Vcc) / Vout_L) - R_fixed;
else
    R_LDR_Left = 1000000; % Cap at 1M if dark
end

if Vout_R > 0.05
    R_LDR_Right = ((R_fixed * Vcc) / Vout_R) - R_fixed;
else
    R_LDR_Right = 1000000;
end

% Increment time step
timeCounter = timeCounter + 1;

% Update Top Graph (Voltages)
addpoints(lineVoltLeft, timeCounter, Vout_L);
addpoints(lineVoltRight, timeCounter, Vout_R);

% Update Bottom Graph (Resistances)
addpoints(lineResLeft, timeCounter, R_LDR_Left);
addpoints(lineResRight, timeCounter, R_LDR_Right);

% Scroll the x-axis for BOTH subplots simultaneously
for g = 1:2
    subplot(2, 1, g);
    if timeCounter > maxPoints
        xlim([timeCounter - maxPoints, timeCounter]);
    else
        xlim([1, maxPoints]);
    end
end

% Render all updates to screen immediately
drawnow;

```

As the microcontroller runs, the data will be streamed in MATLAB as two graphs like so:



Figure 9: An example of voltage and resistance graphs over time.

Conclusion

In conclusion, this project successfully demonstrated the fundamental working principles of LDR and its role in a real-life application. It also highlighted that raw LDR data is rarely immediately usable; rather, it requires careful signal conditioning using the right value resistors, to achieve high sensitivity. By building a closed loop system, we were also able to digitalize and monitor our data in real time, showcasing one of many possibilities of utilizing the LDR.

Appendix

- [1] <https://cdn.sparkfun.com/datasheets/Sensors/LightImaging/SEN-09088.pdf>
- [2] <https://docs.arduino.cc/resources/datasheets/A000005-datasheet.pdf>
- [3] https://www.handsontec.com/dataspecs/motor_fan/MG996R.pdf
- [4] <https://mikroelectron.com/product/me-13392>